

**3.23** The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

- a. Using the quadratic regression model,  $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$ , test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test  $p$ -value. What do you conclude?
- b. Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test  $p$ -value. What do you conclude?
- c. Using the quadratic regression model in part (a), estimate the expected price  $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$  for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.
- d. Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with  $SQFT = 20$  compatible with the result in part (c)? Explain.

a.  $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$

$$E[PRICE|SQFT] = \alpha_1 + \alpha_2 SQFT^2$$

$$\frac{dE[PRICE|SQFT]}{dSQFT} = 2\alpha_2 SQFT$$

$SQFT = 20$  (2000 square feet) and  $SQFT$  increases by 1 (100 square feet)

$$\Rightarrow \text{marginal effect, } ME(SQFT=20) = \frac{dE[PRICE|SQFT]}{dSQFT} \Big|_{SQFT=20} = 2 \cdot \alpha_2 \cdot 20 = 40\alpha_2$$

$$H_0: 40\alpha_2 \leq 13 \text{ (\$13,000)}$$

$$H_A: 40\alpha_2 > 13$$

$$\alpha = 0.05$$

$$\text{test statistic: } t = \frac{\hat{\alpha}_2 - \frac{13}{40}}{se(\hat{\alpha}_2)} \overset{H_0}{\sim} t(500-2)$$

$$RR = \{t \geq t_{0.05, 498} \approx 1.648\}$$

$$t_0 = -26.72875$$

$$p\text{-value} = P(t \geq -26.72875 | H_0 \text{ True}) \approx 1$$

$$p\text{-value} = 1 > 0.05, \text{ do not reject } H_0$$

We have sufficient evidence to indicate that the marginal effect on expected house price of increasing the size of a 2000 square feet house by 100 square feet is not significantly greater than \$13,000 at the 95% confidence level.

b.

$$ME(SQFT=40) = \frac{dE[PRICE|SQFT]}{dSQFT} \Big|_{SQFT=40} = 2 \cdot \alpha_2 \cdot 40 = 80\alpha_2$$

$$H_0: 80\alpha_2 \leq 13 \text{ (\$13,000)}$$

$$H_A: 80\alpha_2 > 13$$

$$\alpha = 0.05$$

$$\text{test statistic: } t = \frac{\hat{\alpha}_2 - \frac{13}{80}}{se(\hat{\alpha}_2)} \overset{H_0}{\sim} t(500-2)$$

$$RR = \{t \geq t_{0.05, 498} \approx 1.648\}$$

$$t_0 = 4.189467$$

$$p\text{-value} = P(t \geq 4.189467 | H_0 \text{ True}) = 1.65395 \times 10^{-5}$$

$$p\text{-value} \leq 0.05, \text{ reject } H_0$$

We have sufficient evidence to indicate that the marginal effect on expected house price of increasing the size of a 4000 square feet house by 100 square feet is significantly greater than \$13,000 at the 95% confidence level.

c.  $E[PRICE|SQFT=20] = 167.3735$  (\$167,373.5)

95% C.I. of  $E[PRICE|SQFT=20] = [158.0481, 176.6988]$

$([\$158,048.1, \$176,698.8])$

Based on the sample and the fitted quadratic regression model, we estimate the average selling price of a 2000 square feet house to be about \$167,373.5

A 95% confidence interval from \$158,048.1 to \$176,698.8 means that the probability that the true average selling price of all 2000 square feet houses in this market falls within this interval is 95%.

d.  $\overline{PRICE}|SQFT=20 = 163.3333$  (\$163,333.3)

$\$163,333.3 \in [\$158,048.1, \$176,698.8]$

Since the sample mean lies within the 95% confidence interval, it is compatible with the result in part c. In other words, the observed average price of the houses with 2000 square feet is consistent with the regression-based estimate of the mean price for the 2000 square feet houses.